

EASY (10 points, 3 questions minimum)

Use tutorial.billboard_top_100_year_end table

Write a query that:

- Shows all top 100 songs from January 1, 1985 through December 31, 1990.

```

2
3 select * from tutorial.billboard_top_100_year_end WHERE year between 1985 and 1990;
4
5

```

	year	year_rank	group_name	artist	song_name	id
1	1985	1	Wham!	Wham!	Careless Whisper	2977
2	1985	2	Madonna	Madonna	Like A Virgin	2978
3	1985	3	Wham!	Wham!	Wake Me Up Before You Go-Go	2979
4	1985	4	Foreigner	Foreigner	I Want To Know What Love Is	2980
5	1985	5	Chaka Khan	Chaka Khan	I Feel For You	2981
6	1985	6	Daryl Hall and John Oates	Daryl Hall	Out Of Touch	2982
7	1985	6	Daryl Hall and John Oates	John Oates	Out Of Touch	2983
8	1985	7	Tears For Fears	Tears For Fears	Everybody Wants To Rule The World	2984
9	1985	8	Dire Straits	Dire Straits	Money For Nothing	2985
10	1985	9	Madonna	Madonna	Crazy For You	2986
11	1985	10	a-ha	a-ha	Take On Me	2987
12	1985	11	Paul Young	Paul Young	Everytime You Go Away	2988
13	1985	12	Phil Collins and Philip Bailey	Phil Collins	Easy Lover	2989
14	1985	12	Phil Collins and Philip Bailey	Philip Bailey	Easy Lover	2990
15	1985	13	REO Speedwagon	REO Speedwagon	Can't Fight This Feeling	2991

Page 1 of 1 Showing rows 1-100 of 100 Columns: 6 Size: 6KB Run: a few seconds

- Returns all rows for songs that were on the charts in 2013 and do not contain the letter "a".

```

1 SELECT * FROM tutorial.billboard_top_100_year_end;
2
3 select * from tutorial.billboard_top_100_year_end where year=2013 and song_name not like "%a%";

```

	year	year_rank	group_name	artist	song_name	id
1	2013	1	Macklemore and Ryan Lewis feat. Wanz	Macklemore	Thrift Shop	6334
2	2013	1	Macklemore and Ryan Lewis feat. Wanz	Ryan Lewis	Thrift Shop	6335
3	2013	1	Macklemore and Ryan Lewis feat. Wanz	Wanz	Thrift Shop	6336
4	2013	2	Robin Thicke feat. T.I. and Pharrell	Robin Thicke	Blurred Lines	6337
5	2013	2	Robin Thicke feat. T.I. and Pharrell	T.I.	Blurred Lines	6338
6	2013	2	Robin Thicke feat. T.I. and Pharrell	Pharrell	Blurred Lines	6339
7	2013	6	Justin Timberlake	Justin Timberlake	Mirrors	6345
8	2013	9	Florida Georgia Line feat. Nelly	Florida Georgia Line	Cruise	6349
9	2013	9	Florida Georgia Line feat. Nelly	Nelly	Cruise	6350
10	2013	12	Lumineers	Lumineers	Ho Hey	6353
11	2013	14	Daft Punk feat. Pharrell Williams	Daft Punk	Get Lucky	6356
12	2013	14	Daft Punk feat. Pharrell Williams	Pharrell Williams	Get Lucky	6357
13	2013	16	Taylor Swift	Taylor Swift	I Knew You Were Trouble	6359
14	2013	21	Anna Kendrick	Anna Kendrick	Cups (Pitch Perfect's When I'm Gone)	6365
15	2013	26	Swedish House Mafia feat. John Martin	Swedish House Mafia	Don't You Worry Child	6373

Page 1 of 1 Showing rows 1-66 of 66 Columns: 6 Size: 5KB Run: a few seconds ago Executed in

- Shows all rows for which T-Pain was a group member, ordered by rank on the charts, from lowest to highest rank (from 100 to 1).

```

8
9 select * from tutorial.billboard_top_100_year_end where artists='T-Pain' order by year_rank desc;
10
11

```

	year	year_rank	group_name	artist	song_name	id
4	2007	93	Chris Brown feat. T-Pain	T-Pain	Kiss Kiss	5627
5	2007	88	Bow Wow feat. T-Pain and Johna Austin	T-Pain	Outta My System	5619
6	2012	86	T-Pain feat. Wiz Khalifa and Lily Allen	T-Pain	5 O'Clock	6316
7	2007	84	Baby Bash feat. T-Pain	T-Pain	Cyclone	5611
8	2008	84	Rick Ross feat. T-Pain	T-Pain	The Boss	5754
9	2008	79	Kanye West feat. T-Pain	T-Pain	Good Life	5748
10	2010	79	DJ Khaled feat. T-Pain, Ludacris, Snoop Dogg and Rick Ross	T-Pain	All I Do Is Win	6024
11	2008	71	Baby Bash feat. T-Pain	T-Pain	Cyclone	5737
12	2007	60	Plies feat. T-Pain	T-Pain	Shawty	5579
13	2006	52	E-40 feat. T-Pain and Kandi Girl	T-Pain	U and Dat	5424
14	2007	49	R. Kelly Or Bow Wow (feat. T.I. and T-Pain)	T-Pain	I'm A Flirt	5563
15	2008	46	T-Pain feat. Lil Wayne	T-Pain	Can't Believe It	5697
16	2006	41	T-Pain feat. Mike Jones	T-Pain	I'm N Lux (Wit a Stripper)	5410
17	2008	40	Lil Wayne feat. T-Pain	T-Pain	Got Money	5689
18	2011	39	Pitbull feat. T-Pain	T-Pain	Hey Baby (Drop It To The Floor)	6112
19	2007	33	T-Pain feat. Akon	T-Pain	Bartender	5538
20	2008	19	Chris Brown feat. T-Pain	T-Pain	Kiss Kiss	5662
21	2009	16	Jamie Foxx feat. T-Pain	T-Pain	Blame It	5799
22	2007	5	T-Pain feat. Yung Joc	T-Pain	Buy U A Drank (Shawty Snappin')	5501
23	2008	1	Flo Rida feat. T-Pain	T-Pain	Low	5639

Showing rows 1-23 of 23 Columns: 7 Size: 707

- Returns songs that ranked between 10 and 20 (inclusive) in 1993, 2003, or 2013. Order the results by year and rank, and leave a comment on each line of the clause to indicate what that line does.

```

10
11 select * from tutorial.billboard_top_100_year_end WHERE YEAR IN (1993,2003,2013) AND year_rank BETWEEN 10 and 20 order by year, year_rank;
12
13
14

```

	year	year_rank	group_name	artist	song_name	id
1	1993	10	Snow	Snow	Informa	3814
2	1993	11	Dr. Dre	Dr. Dre	Nuthin' But A "G" Thang	3815
3	1993	12	Boyz II Men	Boyz II Men	In The Still Of The Nite	3816
4	1993	13	Jade	Jade	Don't Walk Away	3817
5	1993	14	H-Town	H-Town	Knockin' Da Boots	3818
6	1993	15	Jodeci	Jodeci	Latelly	3819
7	1993	16	Duice	Duice	Dazzey Duks	3820
8	1993	17	Robin S.	Robin S.	Show Me Love	3821
9	1993	18	Peabo Bryson and Regina Belle	Peabo Bryson	A Whole New World	3822
10	1993	18	Peabo Bryson and Regina Belle	Regina Belle	A Whole New World	3823
11	1993	19	Janet Jackson	Janet Jackson	If	3824
12	1993	20	SWV	SWV	I'm So Into You	3825
13	2003	10	Evanesence feat. Paul McCoy	Evanesence	Bring Me To Life	4950
14	2003	10	Evanesence feat. Paul McCoy	Paul McCoy	Bring Me To Life	4951
15	2003	11	Lil Jon and The East Side Boyz feat. Ying Yang Twins	Lil Jon and the East Side Bo...	Get Low	4952
16	2003	11	Lil Jon and The East Side Boyz feat. Ying Yang Twins	Ying Yang Twins	Get Low	4953
17	2003	12	Beyonce feat. Sean Paul	Beyonce	Baby Boy	4954
18	2003	12	Beyonce feat. Sean Paul	Sean Paul	Baby Boy	4955
19	2003	13	Nelly, P. Diddy and Murphy Lee	Nelly	Shake Ya Tailfeather	4956
20	2003	13	Nelly, P. Diddy and Murphy Lee	P. Diddy	Shake Ya Tailfeather	4957

Showing rows 1-48 of 48 Columns: 6 Size: 707

MEDIUM (20 points, 2 questions minimum)

Use tutorial.aapl_historical_stock_price table

Write a query that:

- Determines counts of every single column. With these counts, can you tell which column has the most null values?

-Total number of high (3531) column has the most null values.

```

12 -- Determines counts of every single column. With these counts, can you tell which column has the most null values?
13 SELECT
14   count(date) as totalnumber_date,
15   count(year) as totalnumber_year,
16   count(month) as totalnumber_month,
17   count(open) as totalnumber_open,
18   count(high) as totalnumber_high,
19   count(low) as totalnumber_low,
20   count(close) as totalnumber_close,
21   count(volume) as totalnumber_volume,
22   COUNT(ID) AS totalnumber_ID
23 from tutorial.aapl_historical_stock_price;
24
25
26

```

	totalnumber_date	totalnumber_year	totalnumber_month	totalnumber_open	totalnumber_high	totalnumber_low	totalnumber_close	totalnumber_volume	totalnumber_id
1	3555	3555	3555	3541	3531	3535	3555	3547	3555

- Answers the question: What was the highest single-day increase in Apple's share value?

```

25
26 --Answers the question: What was the highest single-day increase in Apple's share value?
27 SELECT date, (high-low)
28 as daily_increase
29 from tutorial.aapl_historical_stock_price
30 WHERE high IS NOT NULL and low is not NULL
31 order by daily_increase desc limit 1;
32
33
34

```

	date	daily_increase
1	5/6/10	59

- Calculates the lowest and highest prices that Apple stock achieved each month.

```

32
33 -- Calculates the lowest and highest prices that Apple stock achieved each month.
34 SELECT year, MONTH,
35 min(low) as lowest_price,
36 MAX(high) as highest_price
37 from tutorial.aapl_historical_stock_price
38 where low is not null
39 and high is not null
40 group by year, month
41 order by year, month;
42
43
44

```

	year	month	lowest_price	highest_price
1	2000	1	21.62	30.38
2	2000	2	24.25	29.98
3	2000	3	28.5	37.59
4	2000	4	26.22	34.88
5	2000	5	20.44	31.56
6	2000	6	20.09	28.81
7	2000	7	23.44	30.91
8	2000	8	22.12	30.75
9	2000	9	12.69	32.06
10	2000	10	8.22	13.38
11	2000	11	8.06	11.5
12	2000	12	6.81	8.75
13	2001	1	7.72	11.25

<< < Page 1 of 1 > >>

HARD (30 points, 1 question minimum)

Write a query that:

- Appends the two crunchbase_investments datasets (including duplicate values). Filter the first dataset to only companies with names that start with the letter "T", and filter the second to companies with names starting with "M" (both not case-sensitive). Only include the company_permalink, company_name, and investor_name columns. Use tutorial.crunchbase_investments_part1 table and tutorial.crunchbase_investments_part2 table

```

50 SELECT
51     company_permalink,
52     company_name,
53     investor_name
54 FROM tutorial.crunchbase_investments_part1
55 WHERE LOWER(company_name) LIKE 't%'
56 UNION ALL
57 SELECT
58     company_permalink,
59     company_name,
60     investor_name
61 FROM tutorial.crunchbase_investments_part2
62 WHERE LOWER(company_name) LIKE 'm%'
63
64

```

	company_permalink	company_name	investor_name
1	/company/tokita-investments	Tokita Investments	01Startups
2	/company/tackk	Tackk	10Xelerator
3	/company/toviefor	ToVieFor	2010 NYU Stern Business Plan Competition
4	/company/trilliant	Trilliant	ABB
5	/company/trilliant	Trilliant	ABB
6	/company/tareaplus	Tareaplus	Academic Partnerships
7	/company/the-virtual-pulp-company	The Virtual Pulp Company	Adtoma
8	/company/tigerspike	Tigerspike	Aegis Media
9	/company/terrajoule	Terrajoule	Air Liquide
10	/company/the-new-motion	The New Motion	Alliander
11	/company/the-new-motion	The New Motion	Alliander
12	/company/the-talk-market	The Talk Market	Amazon
13	/company/the-talk-market	The Talk Market	Amazon

Page 1 of 1

- Displays the number of players in each state, with FR, SO, JR, and SR players in separate columns and another column for the total number of players. Order results such that states with the most players come first. Use benn.college_football_players table

```

65
66
67 SELECT
68     state,
69     COUNT(CASE WHEN year = 'FR' THEN 1 END) AS FR,
70     COUNT(CASE WHEN year = 'SO' THEN 1 END) AS SO,
71     COUNT(CASE WHEN year = 'JR' THEN 1 END) AS JR,
72     COUNT(CASE WHEN year = 'SR' THEN 1 END) AS SR,
73     COUNT(*) AS total_players
74 FROM benn.college_football_players
75 GROUP BY state
76 ORDER BY total_players DESC;
77

```

	state	fr	so	jr	sr	total_players
1	TX	1055	639	649	498	2841
2	FL	944	571	523	497	2535
3	CA	769	478	632	541	2420
4	GA	657	418	368	315	1758
5	OH	443	284	229	207	1163
6	NC	442	248	187	137	1014
7	PA	311	205	189	210	915
8	IL	350	204	185	144	883
9	LA	319	210	178	164	871
10	VA	344	169	176	162	851
11	NJ	232	195	210	187	824
12	AL	299	182	175	112	768
13	--	335	99	102	177	713
14	TN	274	148	150	112	684
15	NY	183	145	138	165	631

Page 1 of 1

BONUS (20 points)

Write a query that:

- Counts the number of companies acquired within 3 years, 5 years, and 10 years of being founded (in 3 separate columns). Include a column for total companies acquired as well. Group by category and limit to only rows with a founding date. Use tutorial.crunchbase_companies_clean_date table and tutorial.crunchbase_acquisitions_clean_date table

```

85 SELECT
86   c.category_code,
87   COUNT(CASE WHEN a.acquired_at_cleaned <= c.founded_at_clean::date + INTERVAL '3 years' THEN 1 END) AS acquired_within_3_years,
88   COUNT(CASE WHEN a.acquired_at_cleaned <= c.founded_at_clean::date + INTERVAL '5 years' THEN 1 END) AS acquired_within_5_years,
89   COUNT(CASE WHEN a.acquired_at_cleaned <= c.founded_at_clean::date + INTERVAL '10 years' THEN 1 END) AS acquired_within_10_years,
90   COUNT(*) AS total_acquired
91 FROM tutorial.crunchbase_companies_clean_date c
92 JOIN tutorial.crunchbase_acquisitions_clean_date a
93   ON c.permlink = a.company_permlink
94 WHERE c.founded_at_clean IS NOT NULL
95 GROUP BY c.category_code
96 ORDER BY total_acquired DESC
97

```

	category_code	acquired_within_3_years	acquired_within_5_years	acquired_within_10_years	total_acquired
1	software	36	72	165	240
2	web	63	90	124	129
3	enterprise	17	38	82	115
4	mobile	22	53	84	109
5	advertising	17	40	63	71
6	games_video	19	42	60	69
7	ecommerce	26	38	50	60
8	biotech	1	7	34	58
9	network_hosting	5	13	30	41
10	hardware	1	7	29	40
11	semiconductor	2	4	25	36
12	security	1	5	20	31
13	social	13	19	29	30
14	analytics	6	14	23	27
15	finance	6	13	17	22

Page 1 of 1 Showing rows